Semantic Search Techniques for Automatic Coding of Economic Activity in Official Statistics

Francesco Ortame 0009-0007-8672-872X, Mauro Bruno 0009-0004-7003-3242

Abstract: This chapter examines embedding-based semantic search as a method to automate the coding of free-text economic activity descriptions into official statistical taxonomies, while respecting the quality requirements of Official Statistics (OS), particularly regarding robust uncertainty quantification. We focus on Sentence Transformers combined with retrieval and optional cross-encoder reranking. Using the Italian ATECO 2022 taxonomy of economic activities as a case study, we evaluate three knowledge base design strategies on a reproducible synthetic dataset of 12,410 user-like queries. Approaches based on creating centroid embeddings that represent ATECO codes achieve the best performance (Hit@5: 92.6% with over 1241 labels), while reranking mainly benefits simpler knowledge-base representations. To meet OS requirements for defensible decisions, we integrate conformal prediction to calibrate similarity scores and provide statistically valid uncertainty estimates.

Keywords: Automatic Classification, Semantic Search, Sentence Transformers, Conformal Prediction, Official Statistics

INTRODUCTION

Over the past decade, the environment in which National Statistical Institutes (NSIs) operate has changed significantly (Dumpert, 2025). Society's digitization, the growth of platform-mediated services, and the rapid advancement of Artificial Intelligence (AI), particularly Large Language Models (LLMs), have transformed how information is produced and accessed. At the same time, data ecosystems have become more complex. Consequently, expectations for Official Statistics (OS) have grown: users increasingly demand more timely releases, greater detail and, most importantly, simpler ways to access and interact with statistical data (European Commission, 2018). Citizens, policymakers, journalists, and researchers increasingly interact with information through conversational systems and search methods that minimize the need for technical skills or specialized vocabulary. Traditional dissemination models in OS may no longer align with the cultural habits of modern users, since they mainly rely on static tables, keyword-based searches, and metadata-driven navigation. When OS are difficult to interpret, their public value is diminished, risking to widen the gap between NSIs and society (United Nations, 2014).

At the same time, the mission of OS is guided by a core set of essential principles: *methodological soundness*, *quality assurance*, *transparency*, *protection of privacy and confidentiality*, and *institutional accountability*. These principles support trust in OS and distinguish NSIs from other information providers. The current challenge is therefore twofold: (1) NSIs must modernize their production and dissemination systems to remain relevant and accessible, while (2) maintaining the trust and rigor that legitimizes their role as authoritative producers of public information. The tension between innovation and trust becomes especially clear in the context of modern AI, where powerful methods also bring opacity and new risks. In this context, NSIs have a responsibility to modernize the way OS are produced and disseminated while preserving their public value, and to frame AI within its core principles, which should be viewed as the foundation that makes innovation reliable and trustworthy rather than as obstacles to it (Ricciato et al., 2019).

Across the statistical production chain, AI, and language models (LMs) in particular, offer opportunities in *data collection*, managing interactions with respondents, assist interviewers and explore new collection strategies, in *data processing*, automating classification and coding tasks, transforming unstructured non-traditional data sources into statistical information, and in *data dissemination*, improving how users access OS, making it easier to retrieve relevant information and supporting users in interacting with statistical content. Importantly, the role of AI in OS is not to replace expert work, but to strengthen it by improving efficiency and accessibility while preserving methodological quality (CCS-UNS, 2017; UNECE, 2025).

Among recent advancements in Machine Learning (ML), the most significant was the introduction of Transformer-based LMs (Vaswani et al., 2017), which demonstrated unprecedented capabilities for representing and generating natural language. Within the approaches enabled by such models, semantic search has emerged as a particularly promising methodology for NSIs. Its value lies in enabling systems that can understand the meaning of user queries expressed in natural language, rather than relying on exact keyword matching. This, in turn, reduces reliance on high-maintenance search and automatic classification systems based on manually curated rule sets. As a result, semantic search has the potential to enhance access to and interaction with OS, provided that quality and trust requirements are adequately ensured (HLG-MOS, 2023).

This chapter contributes to this agenda by examining semantic search as a methodological and operational approach for the modernization of OS. In recent years, the Italian National Institute of Statistics (Istat) has invested in research projects that implement

advanced ML methods and data-driven approaches (Bruno et al., 2025), expanding its effort to include several approaches based on LMs, with the explicit goal of integrating them into production environments in a controlled, feasible, and quality-assured manner. In the following sections, we present a case study based on semantic search methods for the automatic classification of free-text descriptions into a complex official taxonomy of economic activities. We show how this approach can support production workflows by improving the efficiency of classification processes in OS, and how its outputs can be evaluated within statistical frameworks that ensure reliable uncertainty estimates.

Traditional Methods for Automatic Classification into Official Taxonomies

Traditional methods for automated coding in OS rely on string-matching techniques and manually curated rule sets to assign classification codes to textual descriptions, with early systems already explored in the literature (Wenzowski, 1988). Their long-standing use within NSIs reflects the need for transparent and reproducible methodologies.

Within this context, Istat developed CIRCE[1], a general-purpose rule-based system for automated coding across several statistical classifications and languages (D'Orazio & Macchia, 2002; Macchia et al., 2007). CIRCE assigns classification codes by comparing user queries to structured lexical resources. Input texts are first parsed and normalized, then matched against classification-specific vocabularies and synonym lists that encode expert knowledge (Figure 1.1). CIRCE remains extensively used at Istat, primarily to support the assignment of ATECO codes (Istat, 2022), the Italian version of NACE.

[Figure 1.1]

Figure 1.1. *Workflow of the CIRCE system. The process is organized into two main phases: (1) construction of a classification dictionary from the taxonomy via rule-based parsing and text preprocessing, and (2) matching and rule-based scoring of free-text descriptions to produce coding outcomes. (Source: own)*

While rule-based systems like CIRCE offer reliability and interpretability, their reliance on manually curated rule sets introduces scalability constraints, particularly as classifications evolve and linguistic variability in queries increases. These limitations motivate exploring semantic search approaches that reduce manual effort while maintaining quality.

[1] https://www.istat.it/wp-content/uploads/2023/12/Manuale-circe-2-0.pdf

METHODOLOGY

In this section, we explore the use of modern, non-generative LLMs, specifically, Sentence Transformers, to build semantic search tools for automatic classification of user queries based on semantic similarity measures. Then, we describe methods for reliably quantifying the uncertainty of such classifications.

Sentence Transformers and Semantic Search for Automated Coding

Sentence Transformers are language models that map sentences to fixed-size numeric vectors, called sentence embeddings, such that semantically similar sentences are geometrically close in the resulting space (Reimers & Gurevych, 2019). Unlike generative models, they produce a vector representation rather than generated text. This property enables semantic search: instead of relying on rule-based text matching, the most relevant codes for a user query can be retrieved by identifying the nearest embedding in a pre-built index.

Sentence transformers are derived from token-level encoder models such as BERT (Devlin et al., 2019) and multilingual extensions (Conneau et al., 2020), which produce one vector per input token rather than a single sentence-level representation (Figure 2.1). A pooling step, typically averaging all token embeddings (mean pooling) or extracting the representation of a special classification token (CLS pooling), converts these variable-length outputs into a single fixed-size vector. Applied directly to a pretrained BERT model, however, these simple pooling strategies tend to yield sentence representations that are poorly informative, not suited for semantic similarity tasks. Sentence Transformers address this through a dedicated fine-tuning procedure that adapts the embedding space so that similar sentences are mapped to nearby vectors and yields substantially better performance on retrieval and similarity tasks.

[Figure 2.1]

Figure 2.1. *Simplified flow to obtain sentence embeddings from token embeddings via pooling strategies. (Source: own)*

To build a semantic search system for automated coding, we construct a knowledge base, i.e., a structured collection of textual descriptors representing each classification code. In our setting, each ATECO code can be associated with its official title, official description, taxonomy notes, and labeled example queries.

The knowledge base is encoded into embeddings and indexed. User queries are then encoded and matched against the knowledge base via cosine similarity to retrieve the top

candidate codes. We explore three strategies for mapping knowledge base texts to class embeddings, illustrated in Figure 2.2:

a) Concatenating all texts for a class into a string and encoding them into a single embedding (one-to-one mapping).
b) Storing each text separately for each class and aggregating per-text similarity scores at retrieval time (one-to-many mapping).
c) Averaging embeddings of individual texts into a single centroid per class (one-to-one mapping).

Each approach involves trade-offs between compactness, retrieval granularity, and robustness to heterogeneous class descriptions, and the best choice depends on the specific use case.

[Figure 2.2]

Figure 2.2. *Illustration of three different knowledge base creation and mapping strategies. Different shades represent classes in the taxonomy and circles represent sentence embedding vectors. (Source: own)*

As an optional final step, candidate codes retrieved by the sentence transformer (acting as a fast first-stage retriever) can be re-ranked using specialized cross-encoder models, which score each query-candidate pair more precisely. While computationally more expensive, re-ranking can improve the ordering of candidates when accurate top-*k* ranking matters.

Calibrating Uncertainty around Black-box Similarity Scores

Similarity scores produced by sentence transformers are informative but not calibrated probabilities and can exhibit overconfidence, a well-known issue in neural models (Pearce et al., 2021). That is, they can be overconfident or biased for rare classes or out-of-distribution queries. In OS contexts, where decisions must be defensible and error rates controllable, a single score does not provide a robust basis for setting automatic-accept thresholds.

Conformal Prediction (CP) addresses this by wrapping any black-box predictor and producing prediction sets, subsets of possible class labels, with a formal, finite-sample coverage guarantee (Vovk et al., 2005; Angelopoulos & Bates, 2021). Specifically, given a desired miscoverage level α (the statistical confidence level, e.g., $\alpha = 0.10$ for 90% coverage), CP guarantees that the true class will be included in the prediction set at least $1-\alpha$ of the time, regardless of the underlying model. This guarantee holds under the mild assumption that calibration data are exchangeable with test data.

In practice, CP requires a held-out set to compute nonconformity scores – measures of how unexpected the true label is given the model's output – and derives a threshold from their

empirical distribution. Prediction sets are then built for new queries by including all labels whose nonconformity score falls below this threshold. Algorithms such as Adaptive Prediction Sets (APS) and its regularized variant (RAPS) (Romano et al., 2020; Angelopoulos et al., 2020) improve on simpler approaches by producing sets that are small when the model is confident and larger when it is uncertain.

In our setting, CP allows us to define an automatic-accept policy: when the prediction set is a singleton, the system's assignment can be accepted automatically; when the set contains multiple candidates, the item is flagged for manual review. This provides a statistically grounded mechanism for controlling the trade-off between automation and human oversight.

USE CASE: CLASSIFYING USER QUERIES INTO THE OFFICIAL ATECO TAXONOMY VIA SEMANTIC SEARCH

Now that we have explained all the necessary theoretical concepts to build a semantic search classification system, we can present Istat's use case. Specifically, it involves classifying user queries, which contain short descriptions of economic activity, into a specific ATECO code to help them find the right fit within a complex taxonomy.

First, we present the dataset we used and the processing steps we applied. Second, we go through the actual implementation, including Python code for each section. Last, we show preliminary results.

It should be noted that, in the actual use case, we also compute an agreement score between our semantic search-based method and the traditional CIRCE method for a set of real user queries. However, due to privacy constraints, we can't publish that dataset. Therefore, to make this use case publicly available and reproducible, we use a synthetic version that contains queries generated by an LLM, each labelled with the correct ATECO code. This generation process is detailed in the following sections.

Datasets

This use case draws on two complementary datasets that jointly capture economic activity descriptions and their associated ATECO codes. The first dataset is the official ATECO 2022 classification, which provides a hierarchical structure, textual definitions, and additional notes underpinning all coding tasks. The second dataset is a synthetic corpus of user-like queries obtained via GPT-5-mini. We describe both in detail in the following paragraphs.

ATECO Taxonomy

The Italian ATECO taxonomy organizes economic activities into progressively detailed classifications, grouping together entries that share comparable characteristics while adhering to the principles of completeness and full coverage. Each category is strictly exclusive: every activity must be assigned to one (and only one) ATECO code. The categories must also retain practical relevance, meaning the level of detail cannot be so fine-grained as to jeopardize the feasibility of data collection. The structure consists of six hierarchical layers: (1) *Sections*, identified by single letters from A to U; (2) *Divisions*, marked by two-digit codes ranging from 01 to 99; (3) *Groups*, with three-digit codes; (4) *Classes*, with four-digit codes; (5) *Categories*, using five-digit codes; and (6) *Sub-categories*, identified by six-digit codes. Table 3.1 summarizes this hierarchy and the scope associated with each level.

Table 3.1. ATECO 2022 hierarchical structure

	Section	Division	Group	Class	Category	Sub-category
Digits	1	2	3	4	5	6
Size	21	87	270	615	919	1240
Example	A, U	05, 73	46.7, 74.2	46.73	46.73.4	46.73.40

Beyond its system of numeric codes and titles, ATECO is supplemented with a comprehensive set of textual explanatory notes designed to guide accurate classification (Figure 3.1 illustrates the corresponding structure for NACE, the English equivalent of ATECO). These notes serve distinct functions:

- *Central notes* outline the overarching scope and intent of each code.
- *Inclusion notes* supply concrete examples of the activities encompassed within a category.
- *Exclusion notes* delineate the limits of a code by indicating related activities that should be placed elsewhere.
- *Operational notes* offer practical instructions to ensure uniform and consistent application of the classification in real-world contexts.

[Figure 3.1]

Figure 3.1. *Example of a NACE group (01.1) and class (01.11) along with inclusion notes (blue) and exclusion notes (orange). Both inclusion and exclusion notes located in class 01.11 apply to all nested categories and sub-categories. The text has been shortened for clarity purposes. (Source: own)*

The integration of a hierarchical framework with comprehensive explanatory notes makes ATECO both thorough and intricate. While these notes improve clarity and precision for human coders, they also introduce challenges for automated classification: a single enterprise description can correspond to multiple potential codes across different levels of the hierarchy, and selecting the correct code often hinges on small distinctions expressed in natural language. This complexity poses difficulties for traditional string-matching approaches but presents an advantage for contemporary semantic search methods based on Language Models, where rich semantic context can significantly enhance performance.

Synthetic Dataset

To get an idea of our system's performance, we use GPT-5-mini[2] to generate 10 queries for each ATECO sub-category, for a total of 12,410 labelled queries. To obtain the most accurate results, we pass information from the latter as context and include some query examples to help the LLM better match the actual tone and style of real-world user queries, as few-shot prompting often improves performance (Brown et al., 2020). We prompt GPT-5-mini via its API using the following instruction (translated from Italian):

"You are a generator of descriptions of activities carried out by companies and professionals. Here's what you need to do:

- *You receive as input an economic classification (title + details).*
- *You generate 10 examples of very brief descriptions of activities carried out by companies, consistent with the classification.*
- *Do not generate any text that is not one of the descriptions.*

Important information:

- *Focus on the indicated activity; do not generate descriptions with different activities (you may use synonyms).*

Some examples of descriptions:

[6 REAL-WORLD QUERY EXAMPLES]

Format the output as a Python list.

[ATECO CATEGORY TITLE + DESCRIPTION]"

[2] Temperature is set to 1.0. As of writing, OpenAI's API does not allow custom temperatures for GPT-5 models.

We sample 80% of this dataset, balancing by ATECO sub-category, for a total of 9920 queries, to create the knowledge base in scenarios (b) and (c), while the remaining, non-overlapping 20% (2480 queries) is used for evaluation.

Implementation

Now we can detail the processes of knowledge base creation, retrieval and reranking of candidates. We use *BGE-M3* as our multilingual sentence encoder, and the *BGE-reranker-v2-m3*, as out multilingual reranker (Chen et al., 2024).

Knowledge base creation

Here, we explore the three different knowledge base creation methods explained in the previous sections. In particular, (a) a *concatenation* method, where we put all the available information about an ATECO sub-category into an individual string, which we call *descriptor*, and encode it into a single vector; (b) a *separation* method, where we keep all the information about an ATECO sub-category separate and encode each text into its own vector; (c) a *centroid* method, where we encode each text separately and compute the ATECO code-wise centroid averaging all the embeddings.

(a) *Concatenated descriptor per ATECO sub-category (one-to-one mapping)*. In this scenario, we extract all relevant information from the ATECO classification and concatenate it into a single string (*descriptor*), which we then encode with a Sentence Transformer. This results in a knowledge base with 1241 embeddings, each corresponding to an ATECO sub-category.

(b) *Store and encode each text separately (one-to-many mapping).* To create a knowledge base with a one-to-many mapping between ATECO codes and representative texts, we simply encode each representative text separately and store its corresponding ATECO code. We have data from both the official taxonomy and synthetic queries, and we concatenate them to create a single dataset. Each row contains a single text along its assigned code, and the knowledge base is created exactly as in (a).

This results in a knowledge base with 14,724 embeddings, meaning that each ATECO code is, on average, represented by approximately 12 separate texts. In the results, we analyze the effect of including synthetic queries in the knowledge base compared to keeping only the official texts. For simplicity, we use a "max score" retrieval strategy to assign individual scores to ATECO sub-categories.

(c) *Centroid class embedding (one-to-one mapping)*. The centroid-based approach at knowledge base creation is slightly trickier, as we first need to encode each text

separately and only then average the embeddings by ATECO sub-category. Like (a), this results in a knowledge base with 1241 embeddings, each corresponding to an ATECO sub-category.

Performance

Here, we show the results of our experiment. Specifically, we compare basic retrieval via semantic search with further reranking. It should be noted that, in case (a) and (c), reranking is performed via a pairwise comparison between the user query and the descriptors from the top 5 extracted ATECO sub-categories, while in case (b), reranking is performed via a pairwise comparison between the user query and the extracted most similar individual text for each of the top 5 extracted ATECO sub-categories.

Table 3.2 presents the results for each approach using Hit@k metrics, which measure the number of times the correct ATECO code appears among the top k results.

Table 3.2. Retrieval and reranking performance

Metric	Knowledge Base	Retrieval	Reranking
Hit@1	Descriptor *(one-to-one)*	49.03%	55.12%
Hit@1	Separated *(one-to-many)*	55.40%	55.12%
Hit@1	Centroid *(one-to-one)*	69.90%	62.57%
Hit@3	Descriptor *(one-to-one)*	68.21%	72.36%
Hit@3	Separated *(one-to-many)*	74.01%	75.66%
Hit@3	Centroid *(one-to-one)*	88.36%	84.97%
Hit@5	Descriptor *(one-to-one)*	75.54%	75.54%
Hit@5	Separated *(one-to-many)*	81.79%	81.79%
Hit@5	Centroid *(one-to-one)*	92.63%	92.63%

The centroid-based approach clearly outperforms both other approaches by a large margin. However, we need to keep in mind that the descriptor approach does not use synthetic data, which, coming from the same distribution as the evaluation data, might inflate the performance. It is also interesting to notice that reranking the results only really improves the performance in the descriptor scenario, while in other scenarios it is indifferent or detrimental. This could stem from the fact that the reranker finds it easier to handle clearly worse candidates compared to better ones.

Conformalizing similarity scores

To obtain a statistically grounded measure of uncertainty for the predicted ATECO codes associated with an input query, we apply a standard split conformal prediction procedure. We do not employ adaptive prediction sets procedures as they tend to underperform or collapse to a single prediction set size when the distribution of nonconformity scores is flat rather than concentrated, which is often the case with cosine similarity scores.

We split our evaluation set into two equal-sized sets: a calibration set to calibrate the empirical quantile and a test set to analyze the procedure's outcome. We set $\alpha = 0.1$ and obtain an empirical coverage of 92.3%, meaning that, on average, the correct label was found inside a prediction set 92.3% of the time. Figure 3.2 shows the distribution of the prediction set size across the entire test set, i.e., the number of candidate labels that the procedure considers relevant.

[Figure 3.2]

Figure 3.2. *Prediction set size distribution over the test set for the split conformal prediction procedure. (Source: own)*

As we can see, most of the mass lies in the 1-9 range, which is a promising result given the high dimensionality of the label space (1241 labels in total). Such procedures are useful as they provide statistically valid coverage guarantees while also calibrating an inclusion threshold, which is fundamental in semantic search systems that aim to aid users to find a correct code in a complex taxonomy, greatly restricting the range of options.

CONCLUSION

This chapter examined semantic search as a practical method based on modern Language Models (LMs) to support production tasks in Official Statistics (OS), focusing on automated coding into official taxonomies. Starting with the institutional goals and constraints that define OS, such as privacy protection, transparency, reproducibility, quality, and accountability, we argued that some Artificial Intelligence (AI) solutions, especially generative models, require greater prudence than others when implemented in National Statistical Institutes (NSIs). In this regard, non-generative embedding-based semantic search is particularly well-suited, as it enhances users' ability to express their information needs in natural language while keeping the retrieval process transparent.

The ATECO case study shows the potential of this method in a field where complexity and language differences are natural. Modern LMs such as Sentence Transformers can support

the identification of candidate codes with high accuracy levels suitable for interactive coding tools, where the aim is often to produce a short, ranked list of candidate codes rather than a fully automated decision. Importantly, the results suggest that performance strongly depends on how the classification is represented in the knowledge base (in particular, naïve descriptors versus separate texts and centroid-based representations). The implementation of Conformal Prediction (CP) techniques in semantic search can lead to the desired outcome of calibrating an inclusion threshold that guarantees marginal coverage with statistical validity and provides a reliable filter to shortlist candidate labels.

The methodology outlined in this chapter can be extended to virtually any use case where free-text descriptions need to be classified within a pre-defined classification or taxonomy, both for dissemination systems and user interfaces, providing easier interaction with OS via natural language queries, and for actual statistical processes, where information needs to be extracted from unstructured data sources such as administrative data.

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